





Early–late asymmetry of the muon density with the UMD

Selection bias, core reconstruction, and how to recover it

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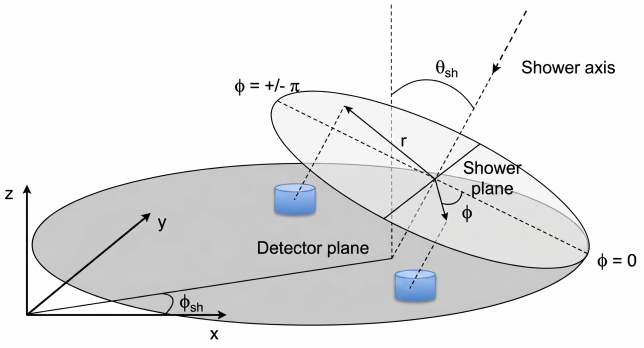
ITeDA (CNEA-CONICET-UNSAM) — FCEN-UBA

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The observable: first harmonic in the shower plane

- ϕ: station azimuth in the **shower plane**.
- Vertical shower: circular symmetry.
- Inclined shower: two distinct regions.
  - **early** ($\phi \approx 0°$): reached first, through *less* atmosphere.
  - **late** ($\phi \approx \pm 180°$): reached later, through *more* atmosphere.
- $A_1 > 0$: early excess; $A_1 < 0$: late excess.
- Fitted on an **ensemble** of showers, per (r, θ) bin.

$$\rho(r, \phi) = \langle \rho(r) \rangle \, (1 + A_1 \cos \phi)$$



Why an early–late asymmetry?

Four mechanisms modulate the muon density in the shower plane:

Mechanism	Effect	Favours
Atmospheric attenuation	The late region is farther: more muons decay on the way	**early**
Detector response	Late muons arrive more inclined: a flat detector (UMD) sees a lower density there	**early**
Kinematic divergence	At the same r, the late region is farther ($1/d^2$ dilution) but needs a smaller opening angle (smaller, more probable p_t)	**early** / **late**
Geomagnetic field	Separates μ^+ and μ^-; second order for $\theta \lesssim 70^\circ$	—

A toy model combining them predicts $A_1 > 0$, growing with r and θ.

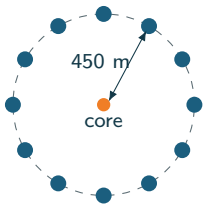
The reference: Monte Carlo geometry

Simulations and the Dense Ring

- Official CORSIKA + *Offline* production, with SD-750 and UMD:

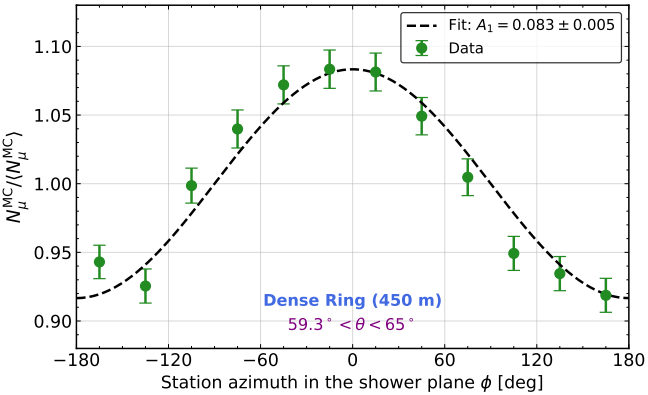
Hadronic models	SIBYLL 2.3e (main), QGSJET-III.01
Primaries	p (main), He, Fe
Energy	$10^{17.5}$–10^{18} eV
Zenith angle	$\theta \leq 65^\circ$

- **Dense Ring**: 12 virtual stations at $r = 450$ m from the **true** axis, every 30° in the shower plane. Same r for all: the radial fall-off of the density cannot produce a modulation in ϕ.



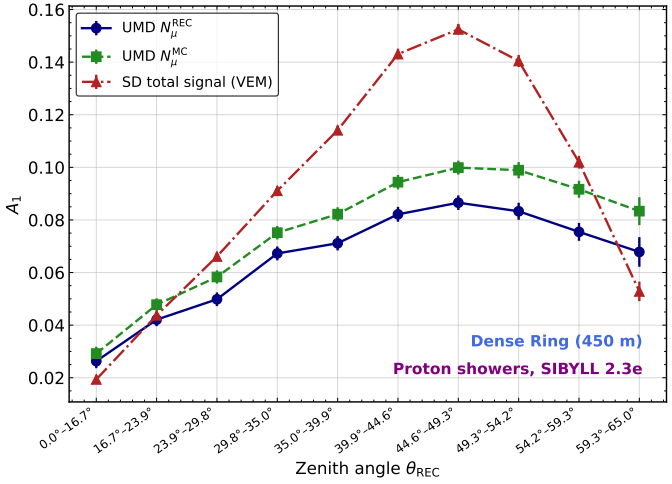
Shower plane.

A clear modulation on the Dense Ring



- Injected muon number N_μ^{MC}, $59.3^\circ < \theta < 65^\circ$.
- Maximum in the **early** region ($\phi \approx 0$).
- Early–late difference $\approx 17\%$ peak to peak ($\pm 8\%$ around the mean).

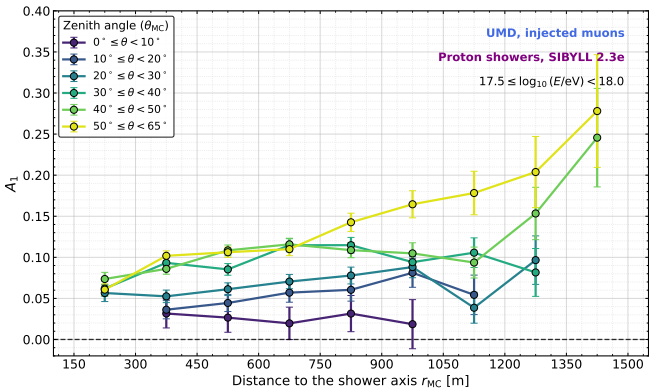
Comparison: SD vs. UMD on the Dense Ring



- UMD: $A_1 > 0$, grows with θ up to $\sim$ 45–50°.
- UMD: lower amplitude with reconstructed (N_μ^{REC}) than with injected (N_μ^{MC}) muons, same shape.
- SD total signal: larger than the UMD at $\sim$ 20–60° (EM-dominated); drops sharply above $\sim$ 50°.
- SD below the UMD only at $\theta \gtrsim 60°$.

UMD radial dependence on the Infill (MC geometry)

Real Infill stations (750 m): the distance to the axis varies from station to station $\Rightarrow A_1(r)$. Station r and ϕ depend on where the axis is: here the **true** core and direction.

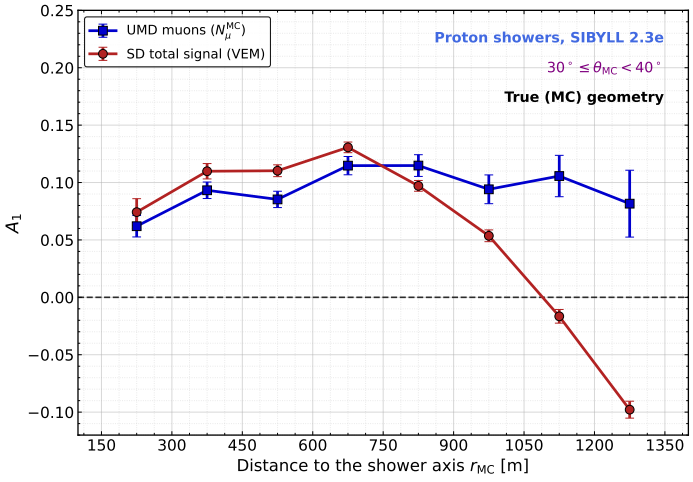


- **UMD only** (injected muons).
- Radial bins of $\Delta r = 150$ m; zenith bins of 10°.
- $\theta < 10^\circ$: compatible with zero.
- 10–40°: grows with r, then **flattens**.
- $\theta \geq 40^\circ$: **grows with** r.

This is our reference.

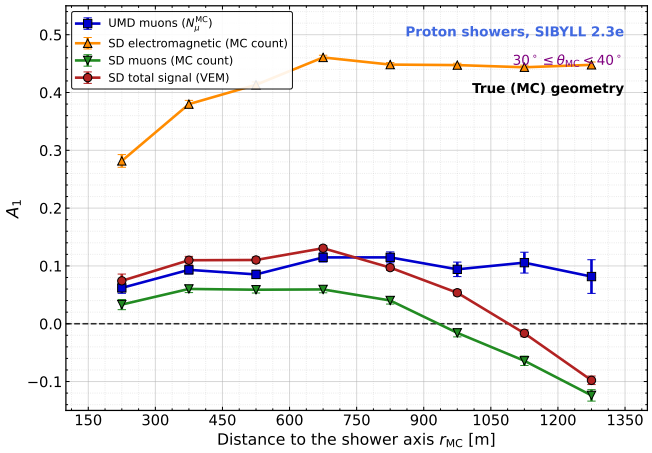
The SD inversion and the selection bias

The SD comparison: inversion far from the core



- We now compare the UMD and SD asymmetries in the same Infill showers.
- MC geometry, $30^\circ \leq \theta < 40^\circ$.
- UMD: positive at all r.
- SD total: a maximum, then negative far from the core.

It is the SD muon component that inverts



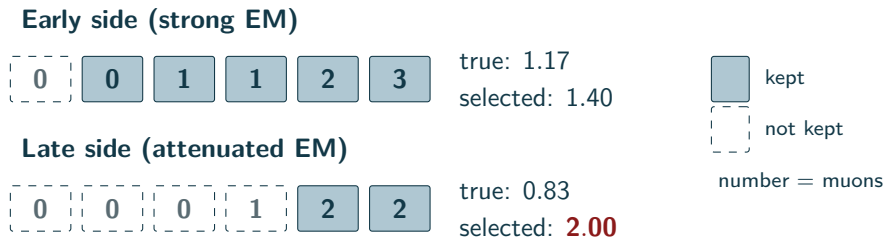
- MC particle counts in the tank, split by component.
- **EM**: positive at all r.
- **SD muons**: positive near the core, negative beyond $\sim$ 900 m.
- The SD total follows its muons at large r.

Same quantity, opposite sign in UMD and SD.

Hypothesis: the station selection biases the average

HYPOTHESIS

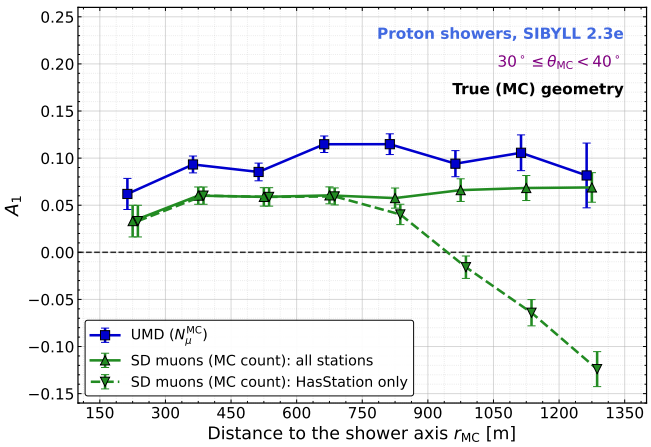
A station enters the analysis only if it **belongs to the reconstructed event** (`HasStation`). If the probability that a station is kept depends on its own muon content, and this dependence is not the same on the early and late sides, the **regional averages are biased**.



Both averages go up, the late one much more: the sign flips **without moving a single muon**.

Illustrative cartoon. In the simulation, at 900–1500 m, 54% of the early and 33% of the late tanks are kept.

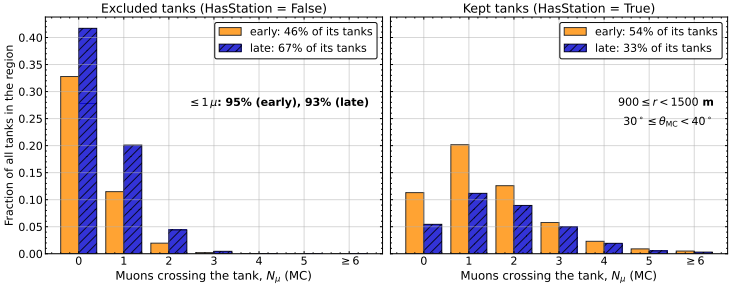
Test 1: without HasStation, the inversion disappears



- Same MC counts, coordinates and fit: only **which stations** enter changes.
- All stations: SD muons **positive at all** r ($+0.03$ to $+0.07$).
- Same sign and trend as the UMD, **but not the same values**.

Test 2: the excluded tanks should hold very few muons

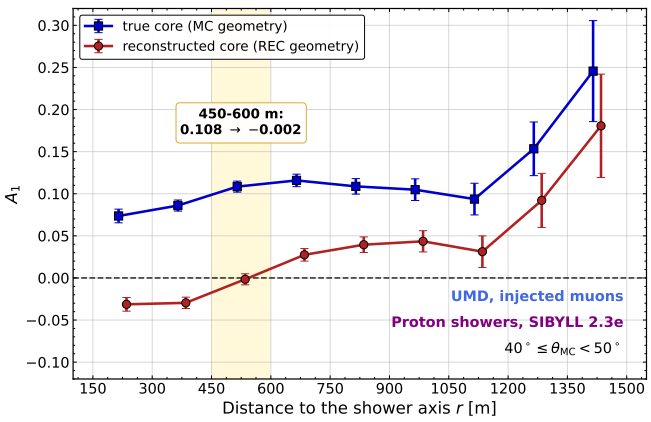
- If the inversion comes from the **trigger**, the tanks dropped by the reconstruction (`HasStation` = False) should hold, on average, at most one muon: too few for the station to trigger.

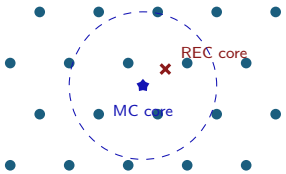


	$\langle N_\mu \rangle$ all tanks	$\langle N_\mu \rangle$ kept tanks	fraction kept
early	0.96	1.49	0.54
late	0.87	1.69	0.33

Reconstructed geometry and the core position

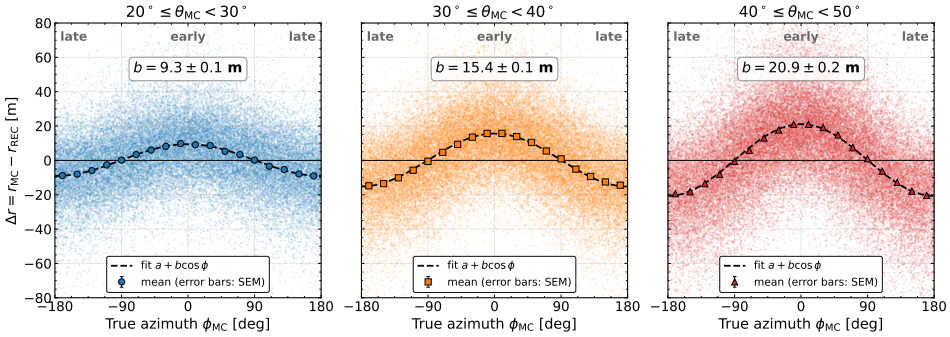
REC geometry suppresses the asymmetry near the core





- Same UMD modules and **injected** muons; **only the core changes**.
- Near the core A_1 drops to **zero or below**; farther out the curves **converge**.
- The effect comes from the **offset between the true and the reconstructed core positions**.

The reconstructed core is shifted toward the early region



- $\Delta r = r_{\mathrm{MC}} - r_{\mathrm{REC}}$ vs. ϕ is fitted by $a + b\cos\phi$, the modulation expected from a **rigid displacement of the core** by b: early stations are placed **closer** to the core, late ones **farther**; $a \simeq 0.2$ m.
- Likely cause: the reconstruction LDF is **symmetric in** ϕ; it absorbs the early excess by **moving the core toward the early side, where there is more signal**.

A shifted core adds a $\cos\phi$ term to the LDF

- Binned by r_{REC}, a station at r is really at $r + b\cos\phi$: its density belongs to **another radius**.

$$\rho(r + b\cos\phi) \simeq \rho(r)\Big[1 - \beta\,\frac{b}{r}\cos\phi\Big] \;\Rightarrow\; \boxed{\Delta A_1 \simeq -\beta\,\frac{b}{r}} \qquad \beta = -\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r}\ \text{(local)}$$

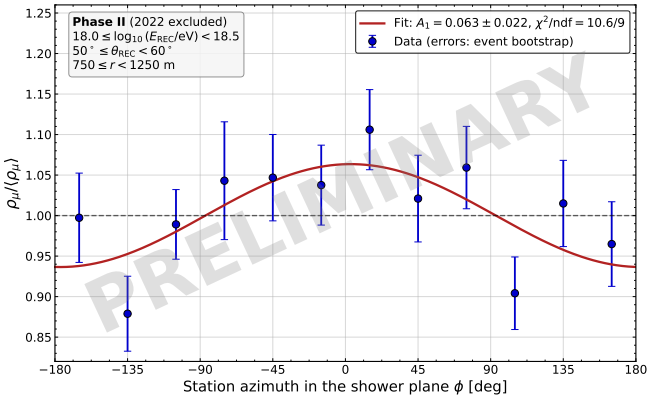
$450 \le r < 600$ m, injected muons

θ	b [m]	β	ΔA_1 measured	$-\beta b/r$ predicted	residual
20–30°	9.3	2.96	−0.050	−0.052	+0.002
30–40°	15.4	2.72	−0.084	−0.080	−0.004
40–50°	20.9	2.52	−0.110	−0.100	−0.010

A first look at field data

The modulation is visible in field data (preliminary)

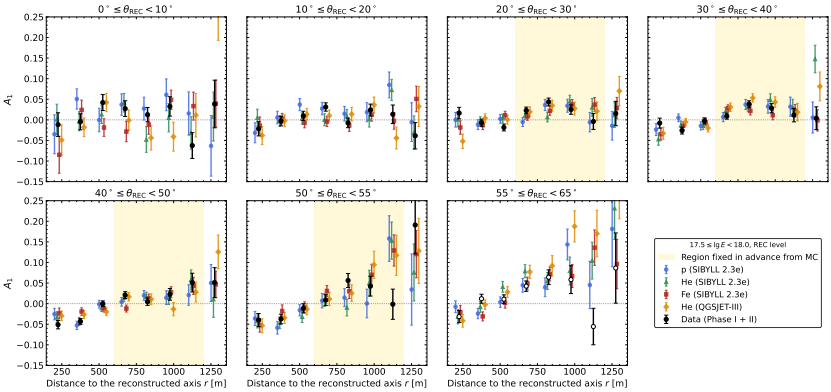
FIELD DATA – PRELIMINARY



- UMD muon density vs. ϕ, normalised to its mean, Phase II; REC geometry.
- Maximum at $\phi \approx 0$: **early excess**, as in the simulations.
- $A_1 = 0.063 \pm 0.022$ (2.8σ), $\chi^2/\mathrm{ndf} = 10.6/9$; errors from a bootstrap over **events**.
- **Preliminary**: no core-shift correction, no systematics, no MC above 10^{18} eV.

Field data follow the REC-geometry simulations

FIELD DATA – PRELIMINARY



- Data (Phase I + II) and simulations with the **same reconstruction**, $17.5 \leq \lg E < 18.0$.
- Same radial pattern: $A_1 \leq 0$ near the core in inclined showers, positive farther out: the **signature of the core shift**, seen in data.

Conclusions

Conclusions

1. In simulations the UMD muon density has a **positive early–late asymmetry**, growing with r for inclined showers.
2. The SD muon asymmetry inverts far from the core **only** when averaging over the stations kept by the reconstruction (a **station-by-station** selection); over all stations it is positive. Excluded tanks hold **mostly** ≤ 1 muon: **consistent with** a trigger effect.
3. With the reconstructed geometry the asymmetry is **suppressed near the core**: the reconstructed core is shifted toward the early region by $b \approx 9$–21 m.
4. The loss follows $\Delta A_1 \simeq -\beta\, b/r$; adding it back recovers the MC reference. Applying it to data requires b **without MC truth**.
5. Field data (preliminary): early excess, $A_1 = 0.063 \pm 0.022$, and a radial pattern that follows the REC-geometry simulations.

Data–MC comparisons need the same station selection and the same geometry.

Next: parametrize $b(\theta)$, $\beta(\theta)$ and validate them on independent simulations; correct the full radial range and field data; separate trigger, readout and reconstruction in `HasStation`.

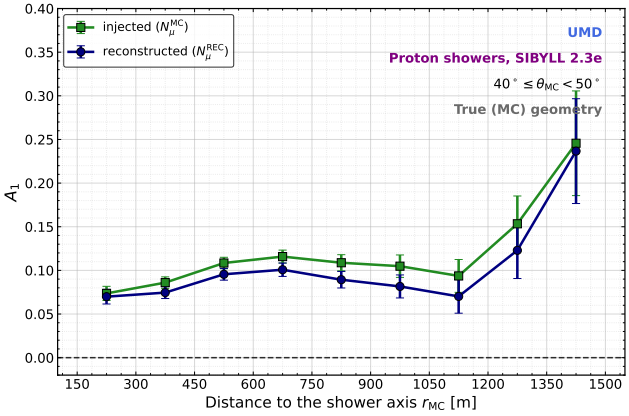


Thank you!

Questions?

Image: A. Chantelauze, S. Staffi, L. Bret (Pierre Auger Observatory)

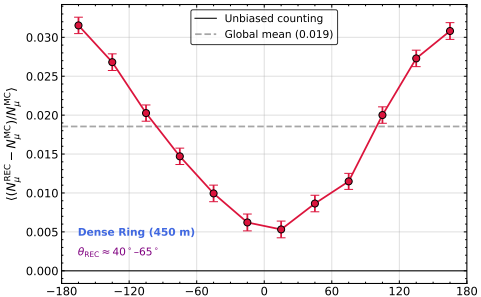
Backup: muon-counting reconstruction (MC geometry)

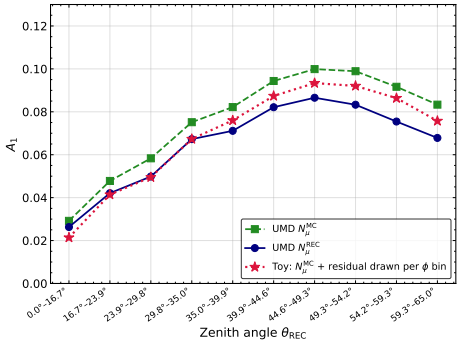


- 40°–50°, true core and direction.
- The reconstructed muon number lowers A_1 by 5–23%, with the same shape.
- A separate effect from the core shift.

Backup: the counting bias depends on azimuth

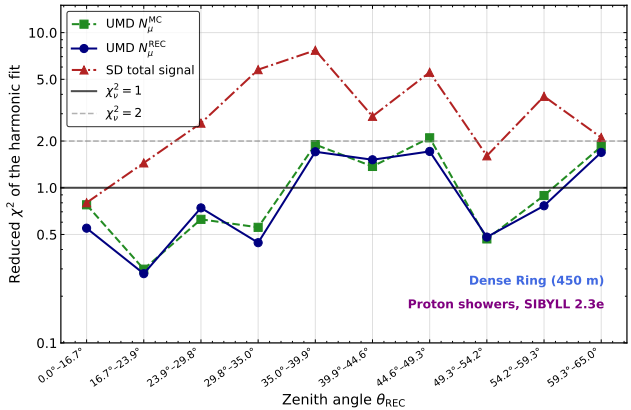
SIMULATIONS – DENSE RING





- Left: mean relative residual $(N^{REC} - N^{MC})/N^{MC}$ vs. ϕ: ≈ 0.005 early, ≈ 0.03 late. More over-counting where there are fewer muons $\Rightarrow$ lower A_1.
- Right: injecting the residuals drawn per ϕ bin into N_μ^{MC} reproduces A_1^{REC} up to $\theta \approx 35^\circ$; above, half to two thirds of the loss. The remnant is open.

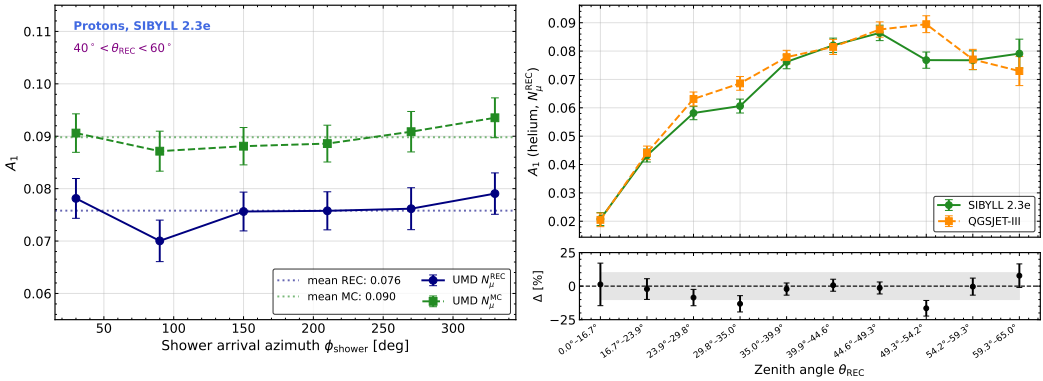
Backup: fit quality on the Dense Ring



- UMD, MC and REC: $\chi^2_\nu \approx 1$ (0.3 to 2.1).
- SD: χ^2_ν up to ~ 8. With such small errors the fit is sensitive to higher harmonics and to correlations between stations of one shower.

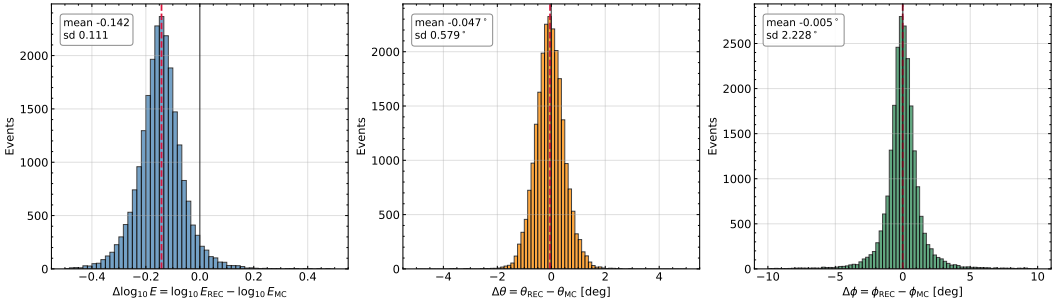
Backup: robustness

SIMULATIONS – DENSE RING



- Left: A_1 does not depend on the arrival azimuth of the shower.
- Right: helium, SIBYLL 2.3e vs. QGSJET-III: within $\pm 10\%$ over most of the range.

Backup: reconstruction performance



- Energy: mean -0.142, spread 0.111 in $\Delta \log_{10} E$ (fewer muons in the simulation than in data, with a data-based energy calibration).
- θ: bias $-0.05°$, spread $0.58°$. ϕ: bias $-0.005°$, spread $2.2°$.

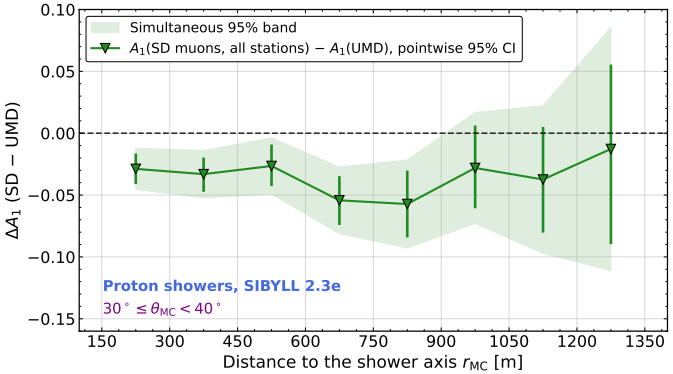
Backup: the core shift and the Dense Ring

- In reconstructed coordinates the ring **is** eccentric, but the stations are physically at 450 m from the **true** axis, and they are not re-binned in r.
- The assigned azimuth rotates by $\sim b/r \approx 2^\circ$, much less than the 30° spacing; the assigned radius changes by at most $b \approx 20$ m.
- Both affect N_μ^{MC} and N_μ^{REC} in the same way, so they cannot explain the MC–REC gap on the ring (that gap is the counting bias).

Backup: is the UMD biased by the selection too?

- The UMD has no trigger of its own: a module is read only if the tank of its station is kept. So the UMD curve is also averaged over `HasStation` stations.
- But the selection acts on the **tank's own particles** (its muons and its EM signal). The module counts **other muons**: a different area, not right under the tank, behind $\sim$2.3 m of soil ($\sim$1 GeV threshold).
- Given the station position, the expected UMD count is still the local muon density: to first order, **no bias**.
- A residual effect can enter through what tank and module share, the fluctuation of the local density from shower to shower and station to station. Not quantified here; the UMD count of the stations that were not kept is not stored in this production.

Backup: SD muons over all stations vs. UMD



- Without `HasStation`, the SD muons stay **below** the UMD: $\Delta A_1 \approx -0.03$ to -0.06 between 150 and 900 m.
- There the pointwise 95% interval excludes zero; beyond 900 m the difference is not resolved.
- Paired bootstrap over the 960 showers (UMD and SD from the same showers).
- Candidates: flat plate vs. tank response, UMD threshold. Open.